

The project record the agent reads

SYNTHETIC FILE	SIMULATES	SYSTEM OF RECORD	WHAT'S INSIDE
01-contract-extract-gc-4.12.md	Conditions of contract + Amendment 2	Contract repo / CLM	GC 4.12 unforeseeable conditions, 28-day bar, Amendment 2 cuts it to 21 days for Areas 6-8
02-geotech-baseline-extract.md	Geotechnical Baseline Report	Document control	Rippable ground baselined to 6.5 m in Area 7
03-daily-reports-area7.md	Contractor daily reports	Aconex	Rock refusal at 2.1-4.5 m, production collapse 14/46 piles
04-correspondence-log.md	Letters + meeting minutes	Aconex / email	C-1187 bare "reservation of rights", MM-088 unclosed action
05-schedule-extract-p6.csv	P6 export (XER)	Primavera P6	Area 7 piling on the critical path, float = 0
06-change-register.csv	Change / variation register	Cost system	Four changes — Area 7 event absent

The point: no single system contains the event. The contract, the ground truth, the letters, the schedule and the register each hold one fragment. Today a human assembles this by hand, if at all.

Seven steps, each with a visible input and output

1 · INGEST

Parse and index every document with provenance

2 · DETECT

Compare field reality against the contractual baseline

3 · MAP CONTRACT

Find the governing clause, including amendments

4 · COMPUTE DEADLINES

Deterministic date math, never LLM-generated

5 · BUILD EVIDENCE

Chain every claim to a cited source line

6 · DRAFT + IMPACT

Notice draft, schedule and cost impact via peer agents

7 · HUMAN GATE

Named approver; nothing leaves without sign-off

Steps 2, 3, 5 and 6 use the LLM to read and reason. **Step 4 is plain code** — a date calculation from extracted anchors, checkable by hand. The split is deliberate: judgment where language is the problem, determinism where being wrong is unrecoverable.

Everything becomes an indexed, citable record

INPUT

6 synthetic files: contract md, GBR extract, 4 daily reports, 5 correspondence items, 7 schedule activities, 4 change records.

```
DR-1042 · 04 Aug 2026 · Area 7 grid  
E12-E15  
Piling rig PR-07 refusal at 2.1 m on  
pile  
group E13. Trial pit confirms cemented  
caprock...
```

03-daily-reports-area7.md



THROUGHPUT

- Chunk by document structure (clause, report entry, letter)
- Extract entities: dates, areas, refs, depths, activity IDs
- Every chunk keeps **file + section provenance** so later claims can cite exactly



OUTPUT

```
{ "doc": "DR-1042", "date": "2026-08-04",  
  "area": "Area 7", "signal": "pile  
refusal",  
  "depth_m": 2.1,  
  "source": "03-daily-reports-  
area7.md#DR-1042" }
```

Indexed record: **37 docs read**, entity graph linking areas, dates and refs.

Field reality vs contractual baseline

INPUT

GBR 5.3.4: "Rippable material is anticipated to a minimum depth of 6.5 m across the Area 7 corridor."

DR-1049: "continuous cemented horizon 2.1-4.5 m across ~60% of the pipe rack corridor... production 14 vs planned 46."

02-geotech-baseline-extract.md • 03-daily-reports-area7.md



THROUGHPUT

- LLM pairs baseline statements with field observations for the same area
- Tests materiality: 2.1-4.5 m vs 6.5 m baseline, 60% of corridor, production collapse
- Checks the change register: **no CR covers Area 7 ground conditions**

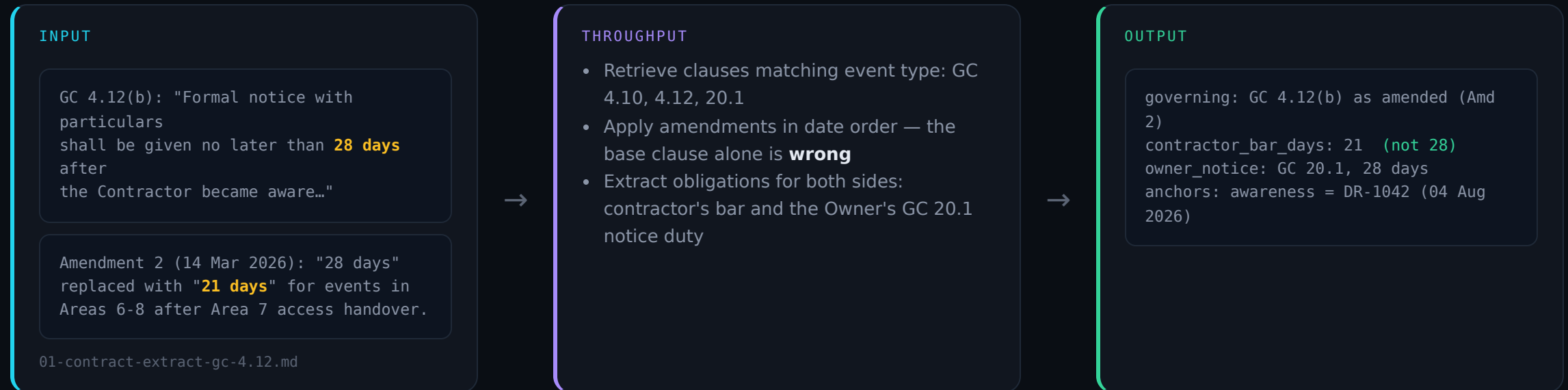


OUTPUT

EVENT EV-047
type: differing_site_conditions
area: Area 7 pipe rack, grids E12-E22
variance: rock at 2.1-4.5 m vs 6.5 m baseline
in_change_register: **false**
confidence: 96%

Why this wasn't caught: the geotech team saw the reports, the commercial team owns the register. The signal crossed a team boundary — exactly where events die today.

Which clause governs, and what did the amendments change?



The amendment catch is the demo moment. A team working from the base contract assumes 28 days. The agent reads Amendment 2 and knows the real bar in Area 7 is 21.

Deterministic date math. The LLM never touches this.

INPUT

Structured anchors from step 3:

```
awareness_date = 2026-08-04 (DR-1042)
contractor_bar = 21 days (GC 4.12(b)
amended)
owner_notice   = 28 days (GC 20.1)
today          = 2026-08-26
```



THROUGHPUT

```
contractor_deadline
  = 2026-08-04 + 21d = 2026-08-25
owner_deadline
  = 2026-08-04 + 28d = 2026-09-01
days_remaining(owner)
  = 2026-09-01 - 2026-08-26 = 6
```

Plain code. Auditable, testable, reproducible.



OUTPUT

```
contractor bar: EXPIRED 25 Aug 2026
owner GC 20.1 deadline: 01 Sep 2026
status: 6 days remaining → ALERT
```

This is the "6 days" banner in the live demo.

Design rule: anything where an error is unrecoverable (dates, sums, deadline math) is computed by code from extracted anchors, with the anchors themselves human-checkable via citations.

Every claim gets a source you can click

INPUT

Event EV-047 + indexed record +
correspondence log.

C-1187 (12 Aug): "We reserve our rights
regarding ground conditions in Area 7...
A detailed submission will follow."
No particulars, no records attached.

04-correspondence-log.md



THROUGHPUT

- Assemble chronology: GBR → DR-1042 → C-1181 → C-1187 → DR-1053 → MM-088
- Judge C-1187 against GC 4.12(b)(i-iii): **bare reservation ≠ notice with particulars**
- Flag MM-088's unassigned "assess contractual position" action

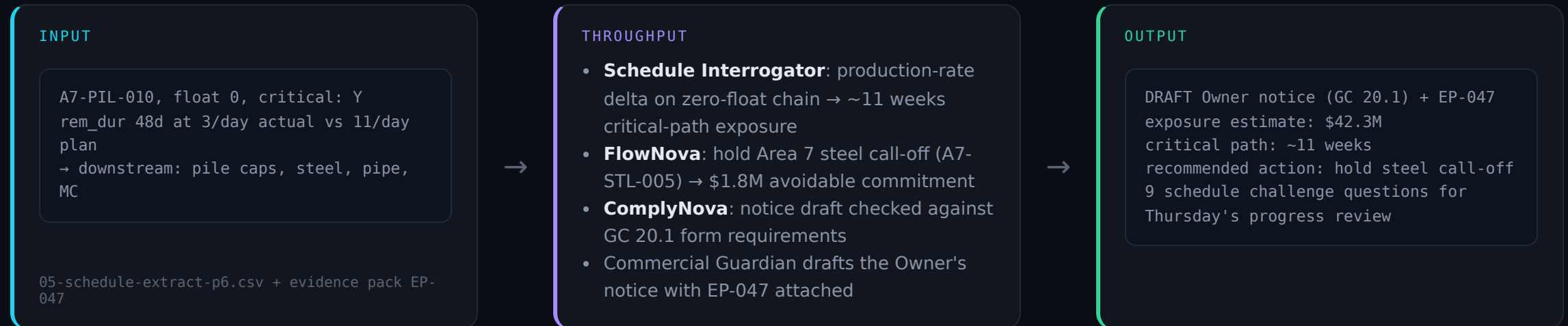


OUTPUT

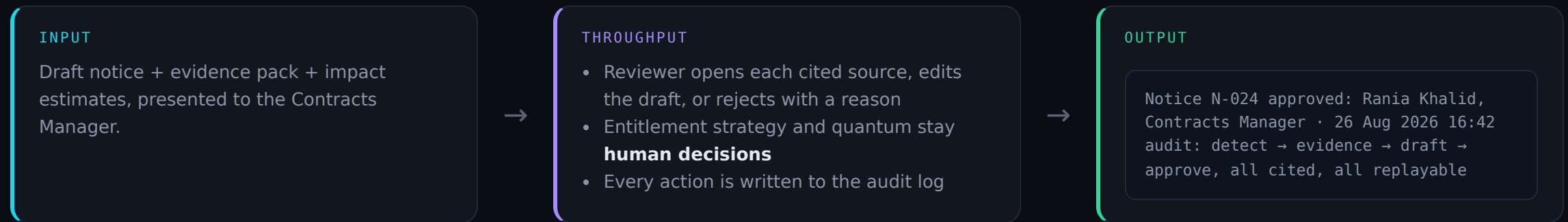
EVIDENCE PACK EP-047 (7 items, cited)

1. GBR 5.3.4 baseline 6.5 m
2. DR-1042 refusal 2.1 m (awareness)
3. DR-1049 60% corridor affected
4. C-1187 insufficient as notice
5. Contractor bar expired 25 Aug
6. A7-PIL-010 float = 0 (critical path)
7. Event absent from change register

Peer agents contribute: schedule, cost, compliance



Nothing leaves the building without a named approver



Afterwards, any stakeholder can ask for a **stewardship brief rendered on demand** from the same record — the throwaway view in the live demo. No dashboard to maintain; the record is the product.

► See it live in the prototype

Input → output, end to end

STEP	INPUT	OUTPUT	WHO DOES THE WORK
1 Ingest	6 raw files, 5 systems	Indexed record with provenance	Parsers + embeddings
2 Detect	GBR baseline + daily reports	EV-047, 96% confidence, not in register	LLM comparison
3 Map contract	GC 4.12 + Amendment 2	Governing clause, 21-day bar (not 28)	LLM retrieval + reading
4 Deadlines	Anchors: 04 Aug, 21d, 28d	Contractor bar expired · owner 6 days left	Deterministic code
5 Evidence	Chronology + C-1187	EP-047, 7 cited items	LLM + citation checks
6 Draft + impact	EP-047 + P6 extract	Notice draft · \$42.3M · 11 wks · hold steel	4 composed agents
7 Human gate	Draft + evidence	Approved notice N-024, audit trail	Named human

Open the files in `synthetic-data/`, follow any output back to its source line, and re-run the date math by hand. **That auditability is the product.**